Pavlos Paparrizos

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Education

B.S. in Mechanical Engineering | University of Maryland, College Park — Expected May 2030

Design Tech High School @ Oracle | Redwood City, California | GPA: 4.32 — June 2026

Featured Project

Efficient RC Plane | Personal Project — February 2024 - Present

- Engineering a 1 m wingspan, sub-250 g aircraft in Onshape targeting 30+ min endurance
- Used slicer and part optimizations to cut mass 18% (263 g to 216 g) in 6 flight-tested revisions
- Developed an open-source OpenFOAM GUI and ran XFoil analyses to optimize the airframe
- Replaced stock LiPo with a custom 2S 18350 Li-ion pack to increase onboard energy by 41%
- Designing custom ESC and FC PCBs in KiCad to reduce wiring and add GPS logging

Experience

Electrical Team Member | *Terps Racing, FSAE EV Team* — September 2026 - Present

- Designing a SolidWorks enclosure for the next-gen 4S LFP LV-system battery pack
- Assembling, testing, and validating the next-gen LV BMS board

Summer Camp Assistant | *Hiller Aviation Museum* — June 2026 - August 2026

- Ran activities for 20+ campers and assisted instructors in a STEM-focused aviation program

Prototype Engineer (Remote) | *Apperture, Swiss Wearables Startup* — September 2025 - August 2026

- Rebuilt 10+ legacy Rhino models of wearable products as parametric, editable Onshape CAD
- Designed toleranced functional parts (bezels, strap clips) for FDM/MSLA manufacturing
- Set up the in-house printer production workflow, handling slicing, printer settings, post-processing, and maintenance across 50+ MSLA batches

Head of Mechanical | *d.Aviation, High School Club* — September 2022 - August 2024

- Built a servo-driven mechanism to pick up and release multiple tennis balls at a time
- Took 3rd out of 12 teams at the 2023 STEM-ED Aero Robotics Competition in San Diego
- Taught members about drone technology and led a 5-person mechanical team
- Repaired operational damage, debugged PX4 firmware, and tuned PID

Skills

CAD & Applications: SolidWorks, Altium, Onshape, KiCad, OpenFOAM

Manufacturing: 3D Printing (FDM & MSLA, 500+ hours), Laser Cutting, Soldering

Programming & Tools: MATLAB, Arduino IDE, Betaflight, Claude Code

Volunteering

Volunteering: 400+ hours at *Samaritan House*, *Fanari Academy*, and *Hiller Aviation Museum* (food distribution, teaching assistance, youth camps)